CSE 3117

Artificial Intelligence

Chapter 1: Introduction

Adapted from slides available in Russell & Norvig's textbook webpage

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Objectives

- To **define** what Artificial Intelligence (AI) is
- To give an overview of **topics** to be covered in this course
- To look at some **applications** of AI
- To get an idea about the **state of the art** in AI

What is AI?

- AI is a relatively new field - started at the end of the 1940s
- The name "Artificial Intelligence" was coined by John McCarthy in 1956
- Two of the many definitions of Artificial Intelligence are:

> "AI is an attempt to understand intelligent entities and to build them" (Russell and Norvig, 1995)

> "AI is the design and study of computer programs that behave intelligently" (Dean, Allen, and Aloimonos,1995)

What is an "intelligent entity" or what does it mean to "behave intelligently"?

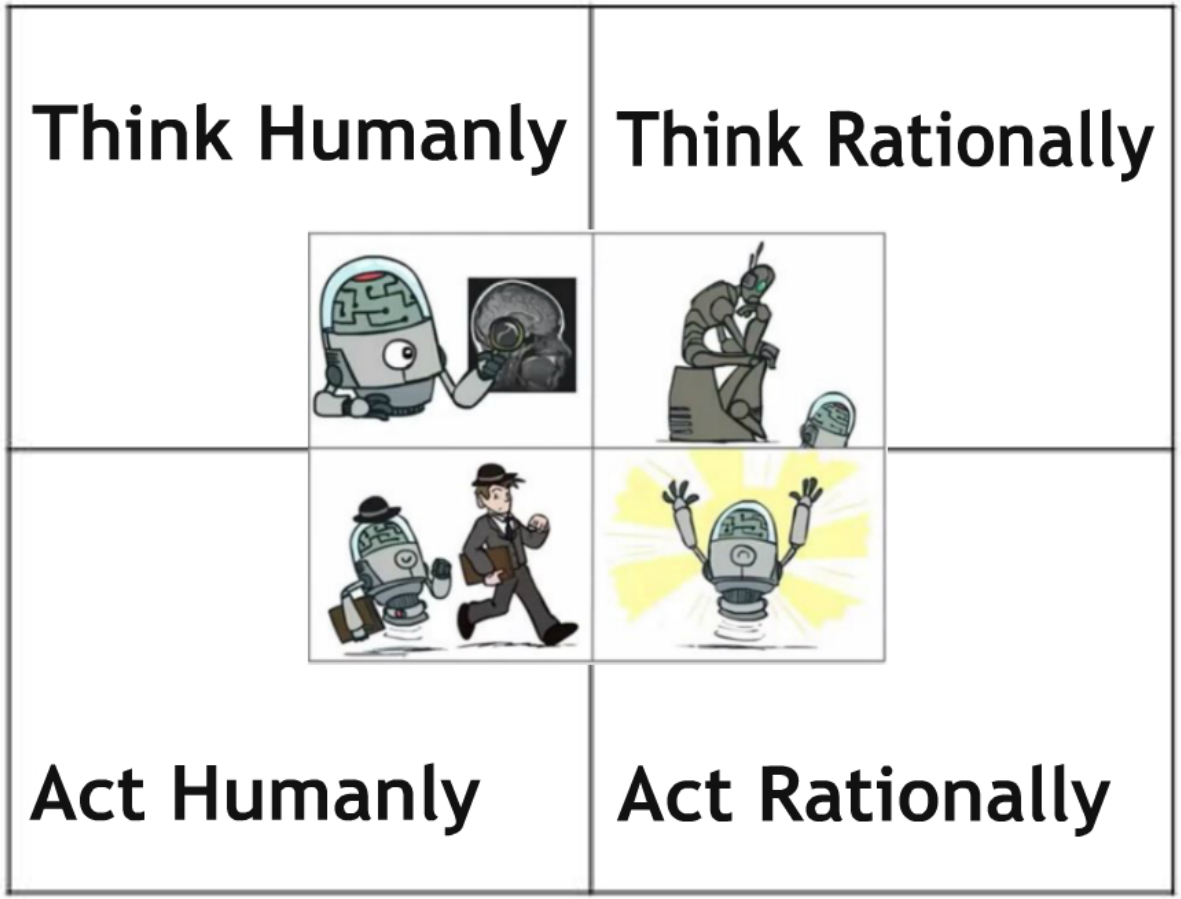
> "The aggregate or global capacity to act purposefully, to think rationally, and to deal effectively with its environment" (Wechsler, 1958)

My favorite definition of AI

> "The study of how to make computers do things at which, at the moment, humans are better" (Rich and Knight, 1991)

What is AI? (cont.)

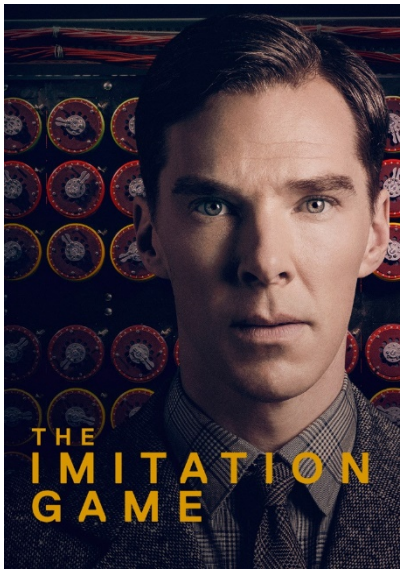
Views of AI fall into four categories:



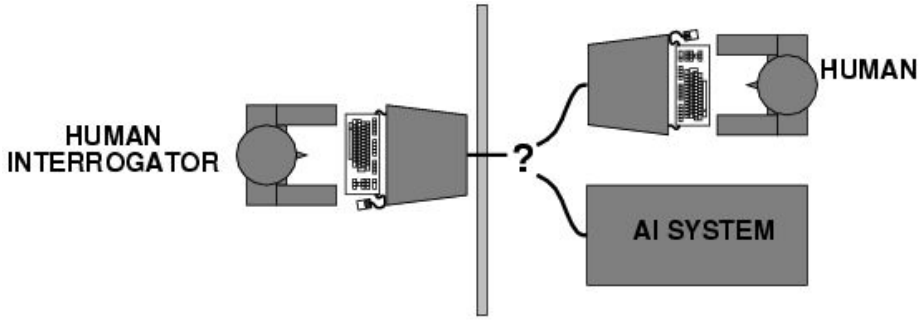
https://kartikkukreja.wordpress.com/2015/05/17/what-is-ai-and-what-can-we-do-with-it-today/

Acting humanly: Turing Test

- Alan Turing (1950) "Computing machinery and intelligence":
- "Can machines think?" "Can machines behave intelligently?“
- Operational test for intelligent behavior: the Imitation Game



The test: A computer is interrogated by a human through a terminal and passes the test if the interrogator cannot tell if there is a computer or a human at the other end.



Acting humanly: Turing Test (cont.)

- To pass the Turing test a machine will need to:
    1. represent knowledge
    2. reason automatically
    3. learn
    4. process natural language
- For the TOTAL Turing test (which includes also a video signal so that the interrogator can test the subject's perceptual abilities) the machine will also need to:
    5. "see" (computer vision)
    6. "move" (robotics)
- Can machines pass the Turing Test?
  - Eugene Goostman (2014): Controversial claim.
  - GPT-4.5 (2025–26): Passed a controlled Turing Test experiment.

Thinking humanly: cognitive modeling

- Requires scientific theories of internal activities of the brain
- Three ways to do this
  - **Introspection** – catching one's own thoughts
  - **Psychological experiments** - predicting and testing behavior of human subjects
  - **Direct identification** - from brain imaging and neurological data
- Cognitive Science brings together computer models form AI and techniques from psychology
- Problem:
  - We know very little about how human brain works!
  - Experimenting on humans is costly and imprecise

Thinking rationally: "laws of thought"

- Aristotle: what are "the right" arguments/thought processes?
- **Syllogisms** provide patterns for argument: always yield correct conclusions when correct premises are given.

  "Socrates is a man. All men are mortal.

  Therefore, Socrates is mortal"
- Logicians in the 19th century developed a notation for statements about all kinds of objects in the real world.
- Problems:
  1. Not all intelligent behavior is mediated by logical deliberation
  2. Difference between solving problems "in principle", and solving them in practice

Acting rationally: rational agent

- Rational behavior: doing the right thing
- The right thing: that which is expected to maximize goal achievement, given the available information
- Doesn't necessarily involve thinking – e.g., blinking reflex – but thinking should be in the service of rational action
  Advantages:
  - More general than the "laws of thought" approach
  - More amenable to scientific development than approaches based on human behavior or human thought

This course will focus on building rational agents

The Foundations of AI

- **Philosophy** Methods of reasoning, mind as a physical system, foundations of knowledge and learning
- **Mathematics** Formal representation and proof, algorithms, computation, (un)decidability, (in)tractability, probability
- **Economics** Formal theory of rational decisions
- **Neuroscience** Physical substrate for mental activity
- **Psychology** Experimental techniques on human and animal behavior
- **Computer engineering** Building fast and efficient computers
- **Linguistics** Knowledge representation, human interaction

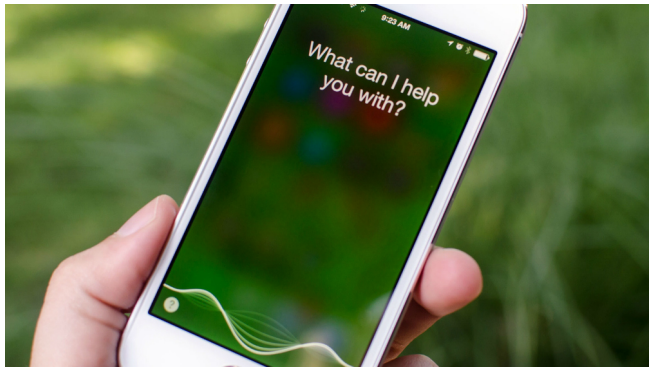
Overview of Topics in this Course

- **Introduction, Intelligent Agents**
- **Problem-solving techniques:**
  - Blind Search, Heuristic Search, Optimal search, Adversarial Search (Game Playing), Constraint Satisfaction
- **Knowledge representation and Logical reasoning**
- **Reasoning with uncertainty**
- **Intro to Machine Learning**
- **Applications of AI**
  - Computer vision, Natural language processing, Robotics

AI in everyday life

Personal and conversational assistants





Recommendation Systems

NETFLIX

amazon

YouTube

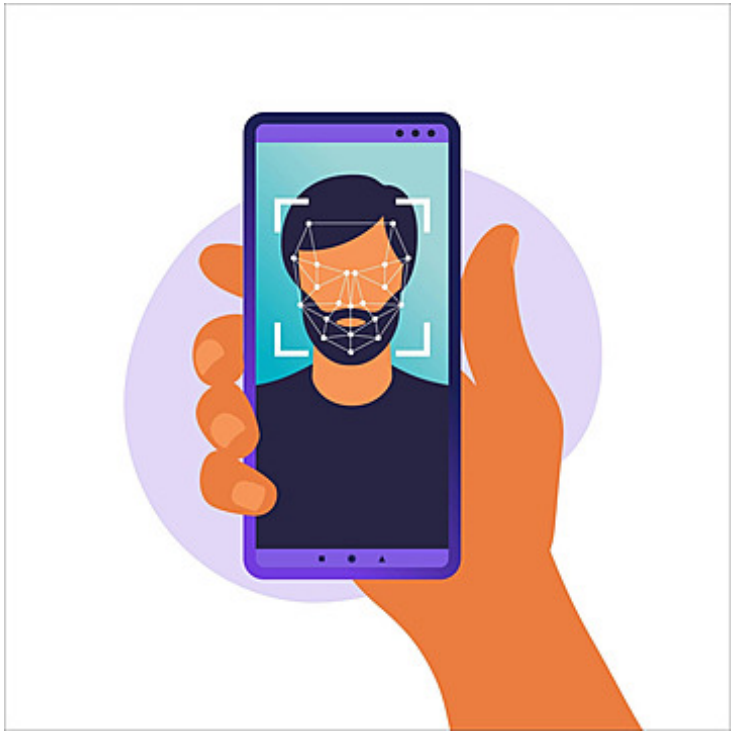
AI in everyday life (cont.)

Email categorization,
Spam filtering



https://sendpulse.com/support/spam-filter

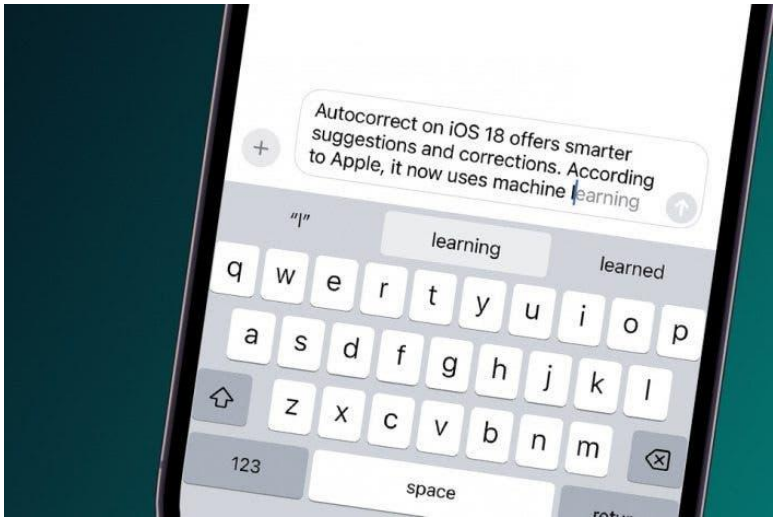
Face recognition



https://pngtree.com/free-png-vectors/face-recognition

AI in everyday life (cont.)

Message autocorrect or prediction



https://www.iphonelife.com/content/autocorrect-iphone

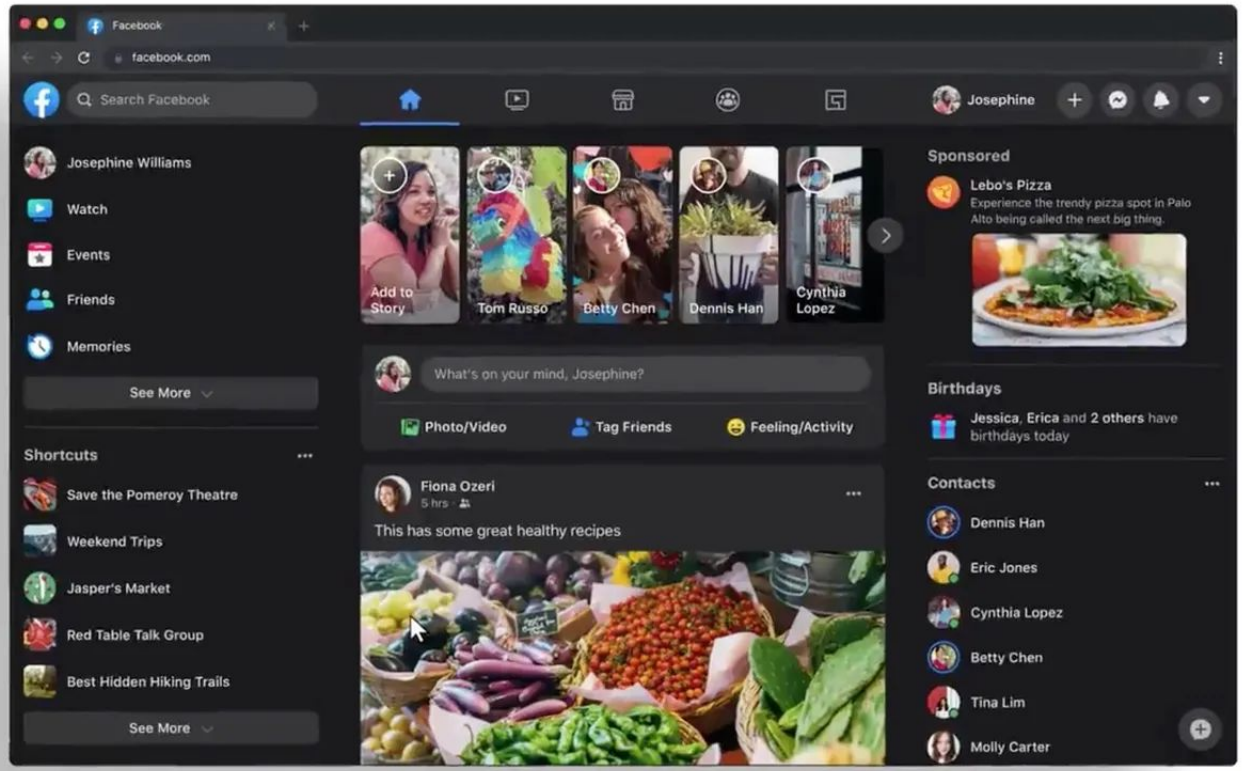
Customer support chatbots



https://www.unisa.edu.au/media-centre/Releases/2023/we-need-to-talk-about-social-chatbots-and-the-impact-of-ai/

AI in everyday life (cont.)

Social Media News Feed and Personalized Ads



https://www.itsnicethat.com/news/facebook-desktop-redesign-digital-200320

AI Success stories

- Game Playing
  - DeepBlue beats Gary Kasparov (1997)

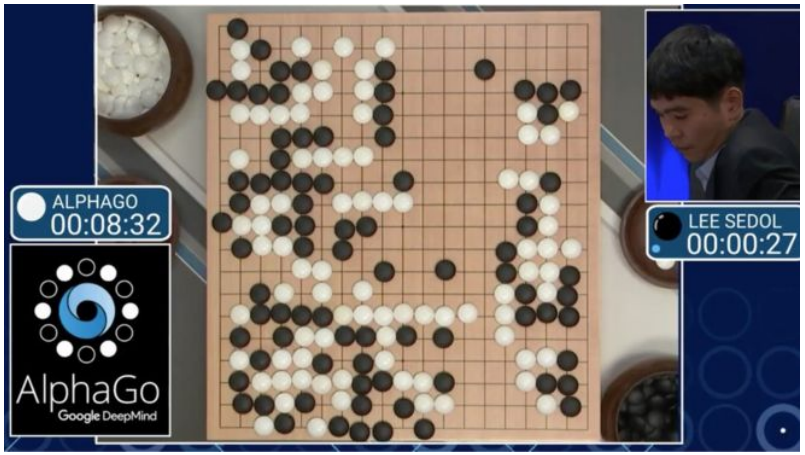


AI Success stories (cont.)

- Game Playing
  - IBM Watson wins Jeopardy! (2011)
  - AlphaGo beats world champion Go player (2016)



https://www.technobuffalo.com/2013/05/21/ibm-watson-smartphone-ap/



http://www.bbc.com/news/technology-35785875

AI Success stories (cont.)

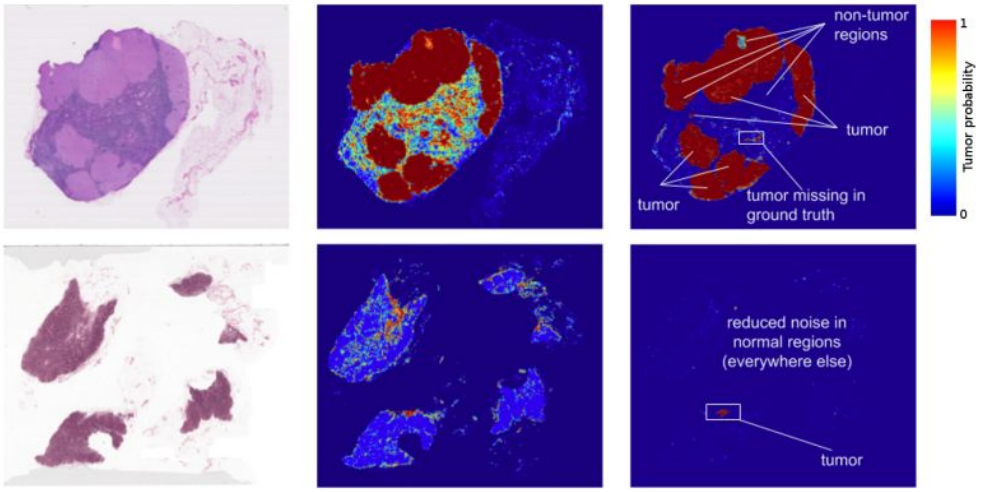
- Autonomous vehicles
  - Google self-driving car, Tesla autopilot

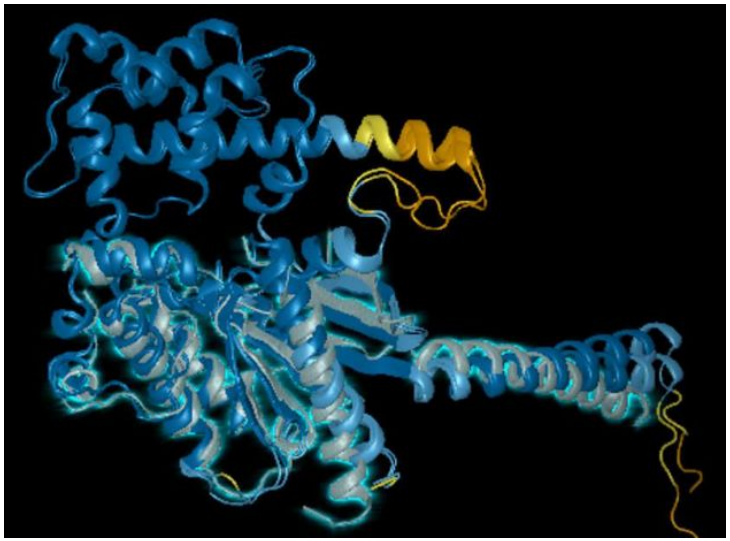


https://www.tesla.com/videos/autopilot-self-driving-hardware-neighborhood-long

AI Success stories (cont.)

- Medical diagnosis
  - AI systems outperformed human experts in blood infection diagnosis and lung cancer diagnosis
  - AI predicts the 3D structures of proteins, contributing to biological and medical research.

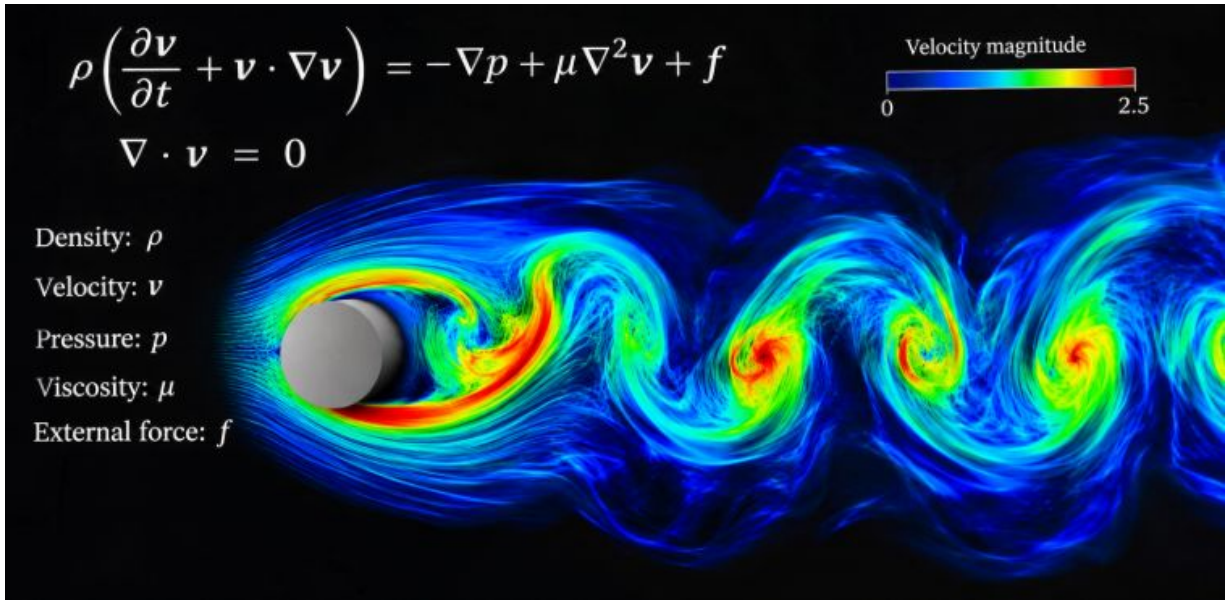




http://wnep.com/2017/03/03/google-uses-ai-to-help-diagnose-breast-cancer/

AI Success stories (cont.)

- Theorem proving & Math problem solving
  - Proved a mathematical conjecture (Robbins conjecture) unsolved for decades
  - AI system have produced a solution to the Navier–Stokes existence and smoothness problem, one of the seven Millennium Prize Problems



AI Success stories (cont.)

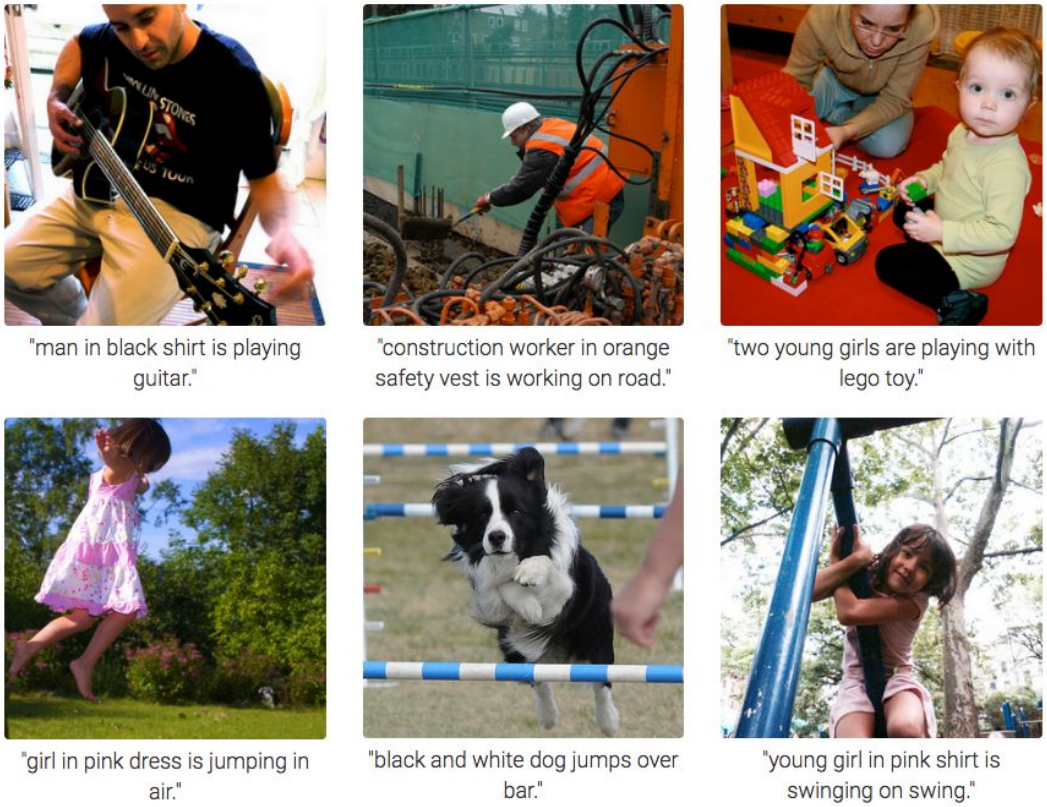
- Robot explorers
  - Mars exploration rovers designed at the NASA Jet Propulsions Laboratory.
  - In 2016, NASA installed a new AI system called AEGIS in the Curiosity rover, enabling it to choose targets autonomously.



https://www.jpl.nasa.gov/missions/mars-science-laboratory-curiosity-rover-msl//

AI Success stories (cont.)

- Image processing
  - Using deep neural networks, possible to identify and categorize objects in images



https://machinelearningmastery.com/inspirational-applications-deep-learning/

AI Success stories (cont.)

- AI generated images
  - Using prompts, AI can synthesize never-before-seen images









Created by MidJourney AI

AI Success stories (cont.)

- Language synthesis

হুমায়ুন আহমেদের মত করে ১০টি বাক্য দিয়ে একটি গরু রচনা লিখ।

গরু খুব শান্ত প্রাণী, কিন্তু তার চোখের ভেতরে একরকম দুঃখ লুকিয়ে থাকে।
ভোরবেলায় যখন সে ঘাস খায়, তখন মনে হয় পৃথিবীটা একটু ভালো।
বৃষ্টির দিনে গরুর গায়ে হাত রাখলে গরম গরম একটা অনুভূতি পাওয়া যায়, অদ্ভুত শান্তি লাগে।
আমাদের বাড়ির গরুর নাম ছিল রুপা, সে সবাইকে চিনত, এমনকি ডাকলেও তাকিয়ে দেখত।
রুপা একবার অসুস্থ হয়েছিল, মা সারারাত তার পাশে বসে ছিল—যেন নিজের মেয়ের মতো।
গরু কখনো রাগ করে না, শুধু চুপচাপ থাকে—এই চুপ থাকাটাই তার সবচেয়ে বড় গুণ।
তার চোখে যে শান্তি, তা অনেক মানুষের মুখে খুঁজে পাওয়া যায় না।
আমি মাঝে মাঝে ভাবি, গরু কথা বলতে পারলে হয়তো অনেক মজার গল্প শুনতাম।
সে হয়তো বলত, মানুষ অকারণে দৌড়ায়, আমরা শুধু ঘাস খাই—তাতেই সুখ।
গরু আসলে এক নীরব ভালোবাসা, যে ভালোবাসা বোঝে সবাই, কিন্তু বলে কেউ না।

Generated by ChatGPT

Final remarks

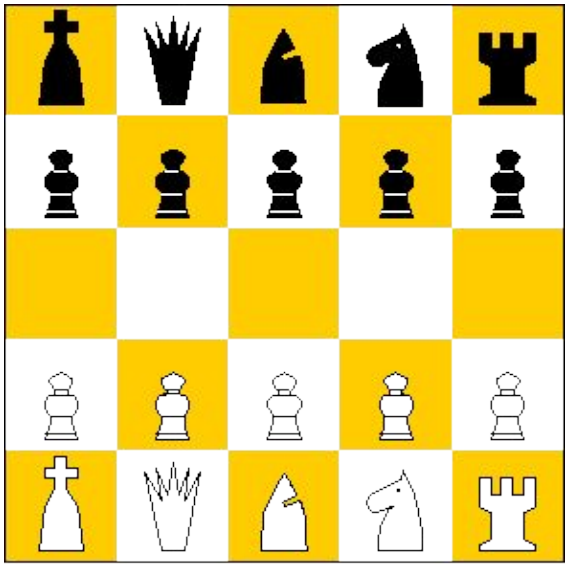


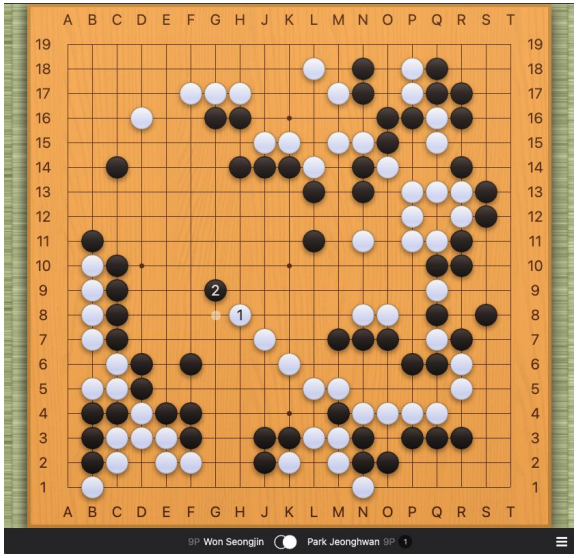
https://twitter.com/mark_riedl/status/837879984375091200

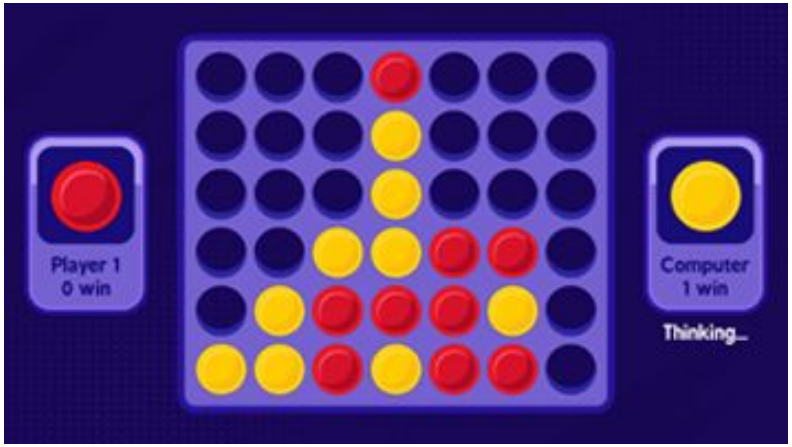
Course Logistics

- Marks Distribution
  - Mid Term 15 marks
  - Projects 25 marks
  - Final 60 marks

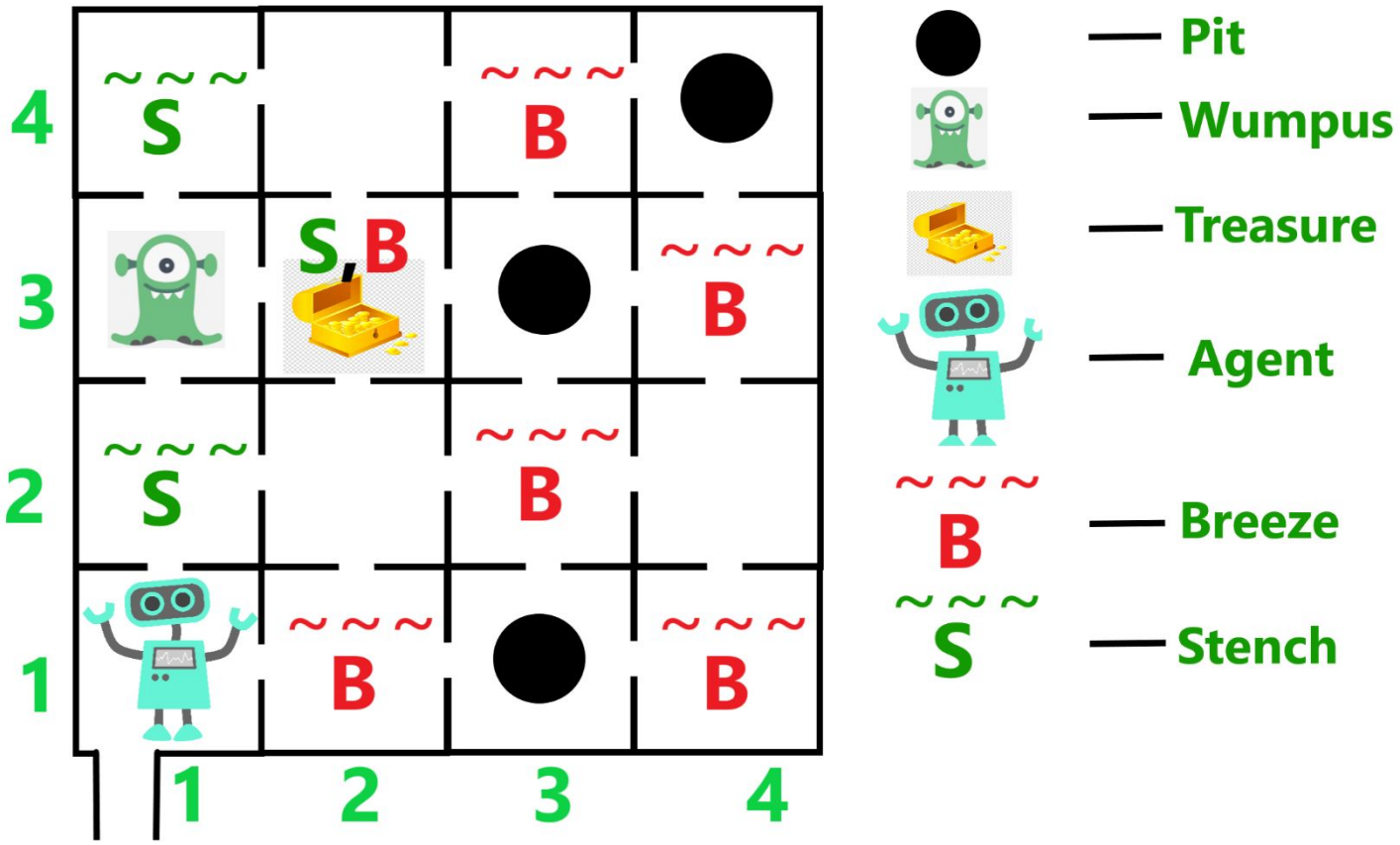
Project 1: AI Game







Project 2: Logic



Project 3: AI for Bangladesh



- Idea generation
- Prototype development
- Poster presentation

https://pressxpress.org/2024/04/20/can-ai-shape-a-fair-safe-future-for-bangladesh/